

Technical Requirement Document

Here's the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data to Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a PySpark job to load customer and order data from CSV files into Delta tables.

Target Cluster Configuration:

- Cluster Name: Data_Load_Cluster
- Databricks Runtime Version: 12.2 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, pyspark

Source Data Details:

- Dataset Name: customerdata
- Location (Path/Table): /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- Format: CSV
- Description: Customer data
- Dataset Name: orderdata
- Location (Path/Table): /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Format: CSV
- Description: Order data

Target Data Details:

- Output Name: customer
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Customer Delta table
- Output Name: order

- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table

Job Flow / Pipeline Stages: Read CSV data, Load into Delta tables

Data Transformations / Business Logic:

- Step: Read customer CSV data and load into Delta table
- Description: Load customer data into Delta table
- Transformation Logic: Use PySpark to read CSV and write to Delta
- Step: Read order CSV data and load into Delta table
- Description: Load order data into Delta table
- Transformation Logic: Use PySpark to read CSV and write to Delta

Error Handling and Logging: Log errors to a designated log file, notify on failure

TR-DLT-002: Transform Order Data by Adding TotalAmount Column

Related Functional Requirement(s): FR-DLT-002

Objective: Implement a PySpark job to add a new column "TotalAmount" to the order Delta table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: order
- Location (Path/Table): gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table

Target Data Details:

- Output Name: order
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table with TotalAmount column

Job Flow / Pipeline Stages: Read order Delta table, Add TotalAmount column, Write to Delta table

Data Transformations / Business Logic:

- Step: Add TotalAmount column to order Delta table
- Description: Calculate TotalAmount by multiplying PricePerUnit and Qty
- Transformation Logic: Use PySpark to add new column with calculated value

Error Handling and Logging: Log errors to a designated log file, notify on failure

TR-DLT-003: Remove Null and Duplicate Records from Customer and Order Tables

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a PySpark job to remove null and duplicate records from customer and order Delta tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location (Path/Table): gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Customer Delta table
- Dataset Name: order
- Location (Path/Table): gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table

Target Data Details:

- Output Name: customer
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Customer Delta table with null and duplicate records removed
- Output Name: order
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table with null and duplicate records removed

Job Flow / Pipeline Stages: Read customer and order Delta tables, Remove null and duplicate records, Write to Delta tables

Data Transformations / Business Logic:

- Step: Remove null records from customer Delta table
- Description: Drop rows with null values
- Transformation Logic: Use PySpark to filter out null records
- Step: Remove duplicate records from customer Delta table
- Description: Drop duplicate rows

- Transformation Logic: Use PySpark to drop duplicates
- Step: Remove null records from order Delta table
- Description: Drop rows with null values
- Transformation Logic: Use PySpark to filter out null records
- Step: Remove duplicate records from order Delta table
- Description: Drop duplicate rows
- Transformation Logic: Use PySpark to drop duplicates

Error Handling and Logging: Log errors to a designated log file, notify on failure

TR-DLT-004: Create ordersummary Table and Load Joined Customer and Order Data

Related Functional Requirement(s): FR-DLT-004

Objective: Implement a PySpark job to create an ordersummary table and load joined customer and order data.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location (Path/Table): gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Customer Delta table
- Dataset Name: order
- Location (Path/Table): gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table

Target Data Details:

- Output Name: ordersummary
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary
- Format: Delta
- Description: ordersummary Delta table

Job Flow / Pipeline Stages: Join customer and order Delta tables, Load into ordersummary table using SCD type 2

Data Transformations / Business Logic:

- Step: Join customer and order Delta tables
- Description: Join on CustId

- Transformation Logic: Use PySpark to join tables
- Step: Load joined data into ordersummary table using SCD type 2
- Description: Maintain history using SCD type 2
- Transformation Logic: Use PySpark to load data with SCD type 2 logic

Error Handling and Logging: Log errors to a designated log file, notify on failure

TR-DLT-005: Update ordersummary Table on Customer Data Change

Related Functional Requirement(s): FR-DLT-005

Objective: Implement a PySpark job to update the ordersummary table on customer data change.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location (Path/Table): gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Customer Delta table

Target Data Details:

- Output Name: ordersummary
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary
- Format: Delta
- Description: ordersummary Delta table

Job Flow / Pipeline Stages: Detect changes in customer Delta table, Update ordersummary table

Data Transformations / Business Logic:

- Step: Detect changes in customer Delta table
- Description: Identify changed records
- Transformation Logic: Use PySpark to detect changes
- Step: Update ordersummary table
- Description: Make old records inactive and new records active
- Transformation Logic: Use PySpark to update ordersummary table with SCD type 2 logic

Error Handling and Logging: Log errors to a designated log file, notify on failure

TR-DLT-006: Create customeraggregatespend Table and Load Aggregated Data

Related Functional Requirement(s): FR-DLT-006

Objective: Implement a PySpark job to create a customeraggregatespend table and load aggregated data.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: ordersummary
- Location (Path/Table): gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary
- Format: Delta
- Description: ordersummary Delta table

Target Data Details:

- Output Name: customeraggregatespend
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customeraggregatespend
- Format: Delta
- Description: customeraggregatespend Delta table

Job Flow / Pipeline Stages: Aggregate data from ordersummary table, Load into customeraggregatespend table

Data Transformations / Business Logic:

- Step: Aggregate TotalAmount column from ordersummary table
- Description: Group by Name and Date
- Transformation Logic: Use PySpark to aggregate data
- Step: Load aggregated data into customeraggregatespend table
- Description: Load aggregated data
- Transformation Logic: Use PySpark to load aggregated data

Error Handling and Logging: Log errors to a designated log file, notify on failure

TR-DLT-007: Implement Data Processing using Delta Live Tables

Related Functional Requirement(s): FR-DLT-007

Objective: Implement the data processing pipeline using Delta Live Tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: Same as previous TRs

Target Data Details: Same as previous TRs

Job Flow / Pipeline Stages: Implement data processing pipeline using Delta Live Tables

Data Transformations / Business Logic: Same as previous TRs

Error Handling and Logging: Log errors to a designated log file, notify on failure